Capstone Dataset and Baseline

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Teach a concrete kit without confusing illustrative scores.

Supply all student rows and a separate teacher key,correct the capstone timeline and record actual review/release status.

Keep one habit: Require source-aware splits,legitimate features and retained baseline evidence before candidate comparison.

MAIN LESSON · SECTION 1

Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

Goal and main teaching plan

Primary target: Define a feasible capstone question and acceptance criteria

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Materials and prerequisite

Teach two 45-minute sessions. This week focuses on capstone data and **requirement**s. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Task, data roles, metric, **baseline**, tie policy and intended use.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Saved classroom project records for critique; no personal student dataset required

Optional reproducible data kit: create IDs i=0 through 71, width=2+(i mod 9) cm, height=2+floor(i/9) cm, and group ID equal to i. Each record is a distinct fictional parcel.

Reference label is blocked when width+height is at least 12, otherwise clear. These are invented operational labels, not real parcel safety labels.

Split by (5*i) mod 18: values 0–9 go to **training**, 10–14 to validation, 15–17 to final test. This gives 40/20/12 rows.

Prepare the kit before class, check both classes in each split, and distribute final-test features without labels. Keep the final key in teacher materials.

The constructed split is not a random sample; students must report that limitation.

Apply the connected idea: Capstone models and baselines

Teach two 45-minute sessions. This week focuses on capstone models and **baseline**s. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

Task, features, group-aware splits, metric, **baseline** and selection policy.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Optional Python 3 editor; no external packages or AI accounts required

Opening retrieval

Without opening the old lesson, explain one key idea from Model Cards and Human Review. Give an example and a mistake that the idea helps you avoid.

Look for: Write a model card whose claims match **evidence**

Reference example, not the only acceptable wording: Teacher explanation: Intended: compare toy classifiers on the documented fictional task.

Not supported: real parcel handling, student assessment or decisions outside the evaluated setup.

Without opening the old lesson, explain one key idea from Subgroup Performance. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate subgroup results with sample sizes

Reference example, not the only acceptable wording: Teacher explanation: A 2/20=10 percent; B 3/5=60 percent; pooled 5/25=20 percent, a 50-point group gap.

Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.

Look for: Read **training**-size learning curves

Reference example, not the only acceptable wording: Teacher explanation: Mean(5+7+6)/3=6; range 7-5=2.

End goal

By the end, I can explain capstone dataset and **baseline** using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Model Cards and Human Review

Subgroup Performance

Learning Curves

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A capstone has two views of each of 40 leaves. Specify a 24/8/8 leaf split and **baseline** procedure. When may final labels be consulted?

Concept to explain

A responsible capstone begins by freezing the problem, intended use, dataset documentation, grouped split, and **baseline evidence**.

Term	Meaning
freeze	lock a reviewed version for comparison
acceptance test	observable required behavior
baseline result	simple method's measured performance
evidence bundle	linked artifacts supporting a claim

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

▼ Teacher-only actual kit final reference key

Teacher-only final ID	Reference label
P03	clear
P07	clear
P14	clear
P21	clear
P25	blocked
P32	blocked
P39	clear
P43	blocked
P50	blocked
P57	blocked
P61	blocked
P68	blocked

Teacher-only actual kit key above belongs to the deterministic synthetic grid specified in this guide. Do not copy it into student kit/programs before the chosen model, feature list, preprocessing, metric/failure policy and membership are frozen. The learner's actual teacher records whether/when release occurred. Public worked examples elsewhere are separate fixtures, not a protected final test for a new project.

MAIN LESSON · SECTION 2

Teach from the actual kit before the separate illustrative results

> Do not score the runnable kit with the unrelated30/10example's expected numbers.

Bounded task and complete actual kit counts

The capstone is a classroom investigation, not a product launch. Choose the supplied fictional clear/blocked parcel task or an equally bounded teacher-approved synthetic task.

Write a question that can be answered by comparing a simple model against a **baseline**. Do not use private student records or real safety decisions.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines capstone data and **requirement**s.

Actual supplied kit role	Rows / distinct parcel groups	Public label availability
Training	40 /40	22 clear, 18 blocked; fit the constant here
Validation	20 /20	9 clear, 11 blocked; score fixed predictions here
Reserved final	12 /12	Features supplied, label cells blank in student script; teacher key held separately
All source IDs	P00–P71; group_idequalsoriginal parcel ID	No source group crosses roles; IDs are audit metadata, not predictive inputs

Actual kit baseline chooses clear from 22 versus18 training labels. It gets 9/20=45%validation accuracy and finds 0/11 blocked cases. Precision is unavailable because it predicts no positives; do not call it perfect. This does not match the separate illustrative baseline 14/20=70%, whose training counts are 30/10. Keep every dataset/version and table labelled.

The kit is a hand-constructed synthetic grid, not a random or representative population sample. The full student program below includes every row; final reference cells are intentionally blank.

Example proposal: can a small classifier improve validation accuracy over the **training**-majority **baseline** on fictional parcel measurements?

Before modelling, assign 40 **training**, 20 validation and 12 final-test records, with related parcel groups kept together.

Define blocked as positive, keep a fixed label rule, and reserve test labels with the teacher until Week 35.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Your teacher can prepare the 72-record fictional grid described in the teacher guide, or approve another documented synthetic dataset. Each supplied record has an ID, group ID, width and height in cm.

Training and validation labels are released for development; final-test labels stay with the teacher until the model is frozen.

The numerical results in later worked examples are illustrative and are not expected scores for your project.

Term	Meaning
Requirement	A concrete condition the project must meet
Acceptance criterion	Evidence that decides whether a requirement is met
Data dictionary	Definitions and units for dataset fields
Milestone	A review point with a stated deliverable

Read separate pictured8-plant plan

Plan the plant capstone around independent groups and a **baseline** before collecting more rows.

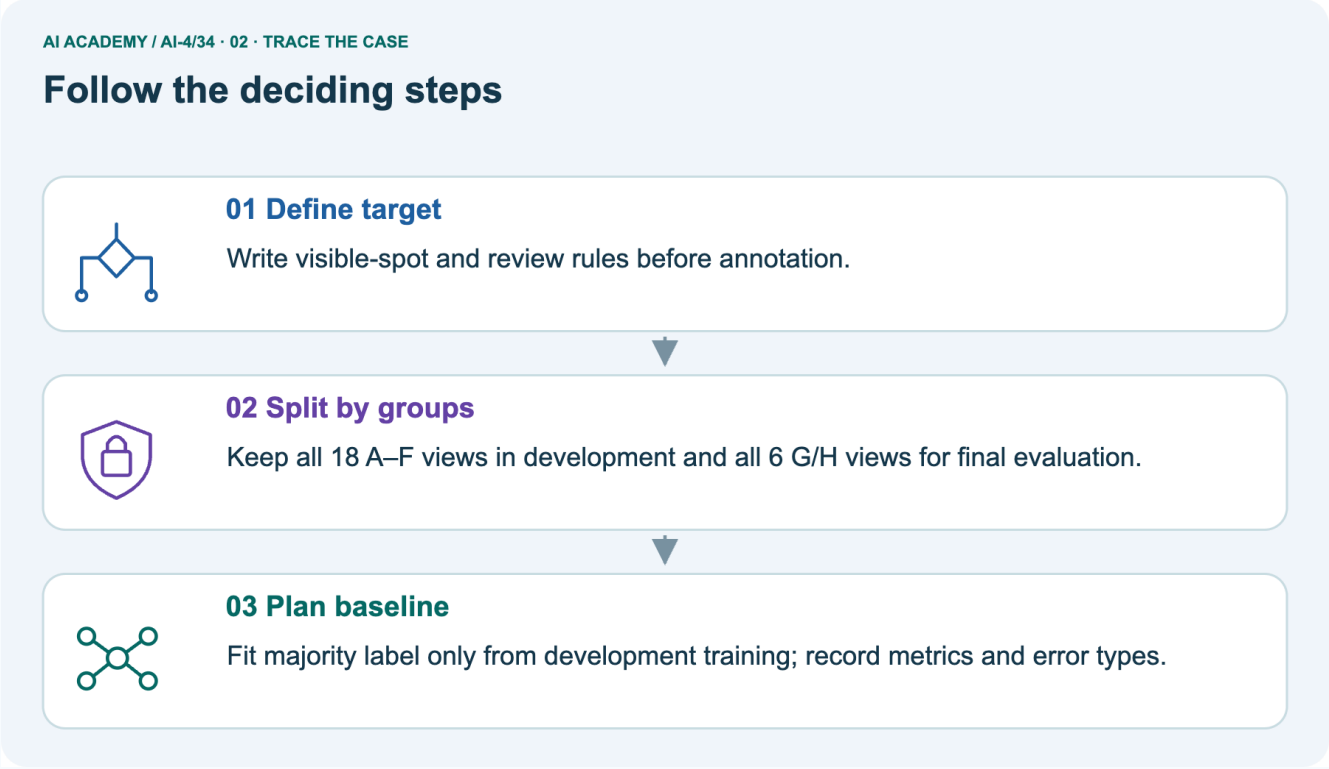
Toy proposed dataset: 8 plants, 3 photos each. Development groupsA–F; final groupsG, H. Target is visible-spot status, with review for hidden leaves.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate small plant plan:8 plants×3 views=24 photos; A–Fdevelopment18 views, G/Hfinal6 views. It is not the 72-parcel kit, whose groups are single-record source parcels. Within plant development, choose and record a further training/validation assignment before fitting; the picture does not supply that assignment.

A derived crop ofGstill belongs to source plantGand cannot enter training whileGis held out. Two final plants provide little population evidence. The actual kit and the independent 40-related-pair task each have their own group/count contracts.

Trace source lineage and crop contamination



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Define target	Write visible-spot and review rules before annotation.
2. Split by groups	Keep all 18 A–F views in development and all 6 G/H views for final evaluation.
3. Plan baseline	Fit majority label only from development training; record metrics and error types.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Fit majority label only from development **training**; record metrics and error types. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

A well-planned split does not create sufficient sample size; two held-out plants support only a very narrow claim.

Change: A derived crop of plantG is added to **training**.

Prediction: The final group separation is violated.

Reason: The crop retains the same source plant identity even with a new file name.

Use the corrected capstone timeline

Capstone data and requirements at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Week 34	Task,data,split and baseline	Teacher checks feasibility
Week 35	Train,validate,analyze and freeze	Record candidate choices
End Week 35 / Week 36	Teacher-gated final check and defense	Evidence plus limitations

The data dictionary lists record ID, group ID, width in cm, height in cm and reference class. IDs track provenance and splitting; they are not automatic predictive inputs. Check uniqueness, units, missing values, class coverage and group separation. Fit imputation or scaling only on training.

Explain why record ID, group ID and measured width have different roles. Explain the steps or evidence behind your response.

▼ Optional illustrative baseline/candidate fixture

Build the **baseline** before the candidate. For classification, use the most common **training** class, with a declared tie policy.

Choose one simple candidate using techniques already learned, such as k-nearest neighbours with **training**-only scaling and a fixed k. Record every development attempt; complexity earns no automatic credit.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines capstone models and **baseline**s.

Illustrative **training** has 30 clear and 10 blocked cases, so the **baseline** always predicts clear. On 20 validation cases with 14 clear and 6 blocked, its matrix is[[14, 0],[6, 0]], accuracy 70 percent.

Candidate matrix[[12, 2],[2, 4]] gives accuracy 80 percent, precision 4/6 and recall 4/6. These are example results, not the student team results.

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

Guided response: Accuracy 16/20=80 percent; precision 4/6 and recall 4/6, both about 66.7 percent; FP 2 FN 2.

Exit response: Model and pre**process**ing version, settings, feature order, split IDs, **prediction**/scoring rules, validation comparison, metric and **decision record**.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION **Freeze** the dataset card, split, and **baseline result**s before **training**.

MODEL IDEA A responsible capstone begins by freezing the problem, intended use, dataset documentation, grouped split, and **baseline evidence**.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD **Freeze** dataset card v3 and grouped split s2 with no plant identity crossing sets. Test set remains sealed; validation supports model choices.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 1 **Freeze** dataset card v3 and grouped split s2 with no plant identity crossing sets.

STEP 2 Test set remains sealed; validation supports model choices.

STEP 3 Validation has 12 healthy and 8 spotted leaves. The always-healthy **baseline** scores 12/20=**60%** accuracy and 0/8=0% spotted recall. The rule has TP=5, FN=3, TN=10, FP=2: accuracy 75% and spotted recall 62.5%.

STEP 4 Acceptance requires valid citations, metric calculations, subgroup review, and safe escalation.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

Capstone has60 independent leaves, several photos per leaf, healthy/spotted labels.

Plan grouping and **baseline**.

Exact explanation to model

Keep all views together in train/validation/test partitions. Define labels/use, fit a majority **baseline** on **training** labels and evaluate declared metrics.

Explain the added worked example

Key terms

Apply the connected idea: Capstone models and baselines

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

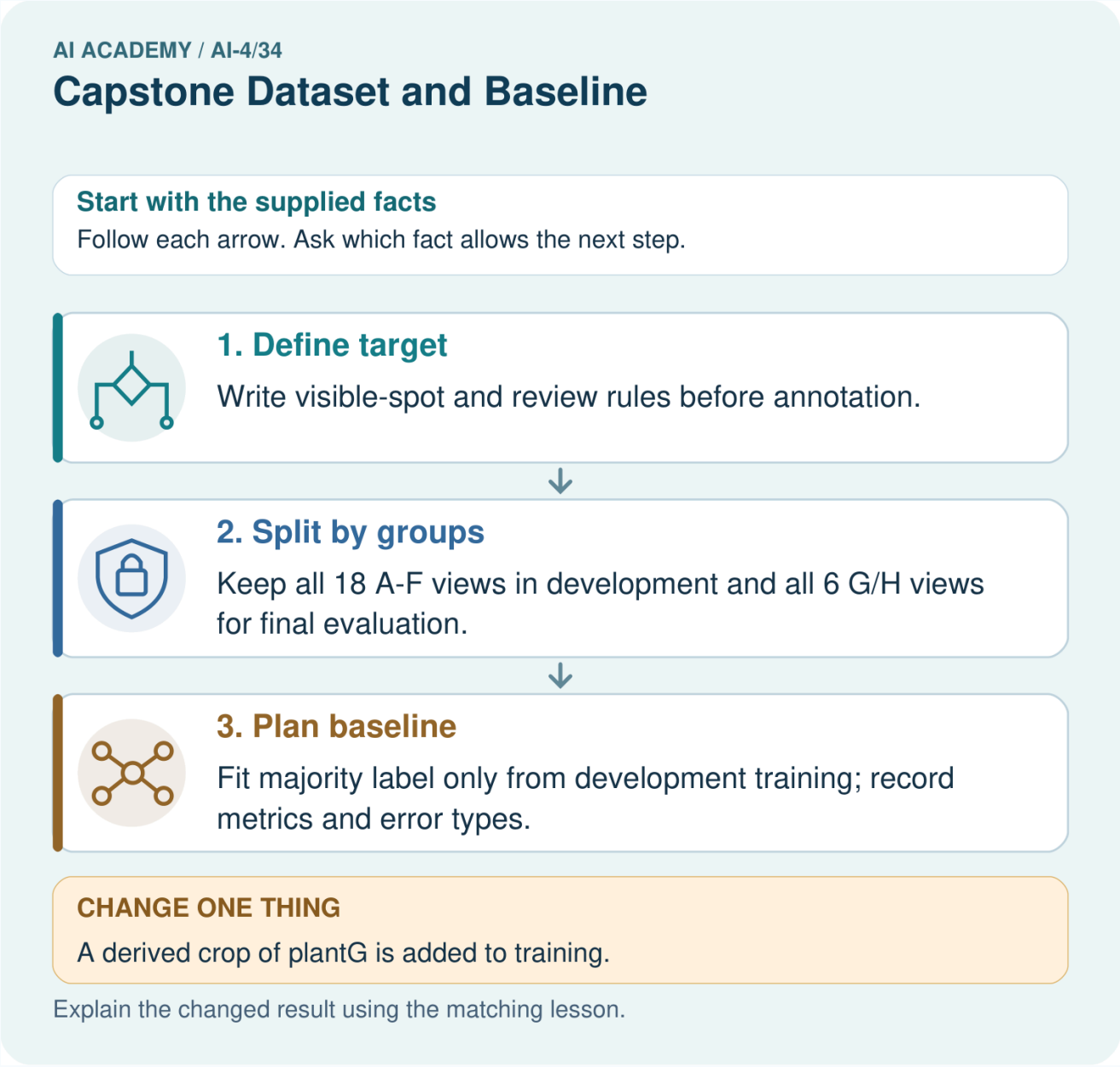
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Capstone Dataset and Baseline

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Guide the calculation, then remove help

> Collect each learner’s prediction before showing its key.

Current completion task and checked key

Completion task: Explain why record ID, group ID and measured width have different roles.

Teacher check: Teacher explanation: Record ID tracks rows, group ID keeps related records together, and width is a potential measured feature available before **prediction**; IDs are not automatic **input**s.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Capstone data and **requirement**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record.

For this week: Compare a simple classifier with a **training**-majority **baseline** on the fictional parcel evaluation; no real handling or student decisions are supported.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

What provenance field prevents the crop from appearing independent? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Store original_plant_id=G and the parent image ID; all derived G records must remain with their assigned final group. Changed-case check: The final group separation is violated.

The crop retains the same source plant identity even with a new file name.

Changed-case reasoning: The final group separation is violated. The crop retains the same source plant identity even with a new file name.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: Capstone has60 independent leaves, several photos per leaf, healthy/spotted labels.

Write the question to explain: Plan grouping and **baseline**.

Reveal after **prediction**s: Keep all views together in train/validation/test partitions. Define labels/use, fit a majority **baseline** on **training** labels and evaluate declared metrics.

Use the earlier transfer question as additional practice. Reserve the selected fresh80-row40-related-pair split repair for the first individual attempt and record support used.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided response: Record ID tracks rows, group ID keeps related records together, and width is a potential measured feature available before **prediction**; IDs are not automatic **input**s.

Exit response: Label the work as development or a demonstration, report the limitation and avoid an untouched-test claim; do not fabricate test data or results.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Use the prerequisite question: Which evaluation choices should precede model selection? Check the explanation before proceeding.
7 to 18 minutes	Model the worked proposal, then introduce the prepared fictional dataset and its data dictionary. Students choose the supplied task or a bounded approved alternative.

Stage	Teacher action
18 to 32 minutes	Teams draft their question, non-use statement and three process acceptance criteria. Use W1–W3 as prompts while preparing the proposal.
32 to 45 minutes	Teams check IDs, units, missing values and split membership. Collect the proposal and identify repairs before any model fitting.
Session 2 start to 8 minutes	Retrieve the distinction between requirement and acceptance criterion. Invite a counterexample to an overconfident claim.
8 to 20 minutes	Complete the independent W4 split critique first. Then teams revise their own data plan and document group separation.
20 to 35 minutes	Review each team proposal and data dictionary for feasibility. Confirm synthetic-data scope, task metric and baseline; reserve final-test labels before approving modelling.
35 to 45 minutes	Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Teach the connected idea

Stage	Teacher action
Session 1 start to 7 minutes	Use the prerequisite question: What must be fixed before the capstone comparison? Check the explanation before proceeding.
7 to 18 minutes	Trace the worked baseline/candidate comparison and check W1–W3 briefly. State that its numbers are illustrative, not target results.
18 to 32 minutes	Teams fit their baseline from training only. Save training class counts, split IDs and baseline validation predictions.
32 to 45 minutes	Teams build one candidate using the earlier k-nearest-neighbour procedure or an approved simple alternative. Check preprocessing and trace one prediction.
Session 2 start to 8 minutes	Retrieve the distinction between baseline and prediction ledger. Invite a counterexample to an overconfident claim.
8 to 20 minutes	Teams evaluate on validation, record the full comparison and explain gains and regressions. Log every attempted setting.
20 to 35 minutes	Collect W4 independently, then review team freeze records. Resolve pipeline failures on development data before requesting final-test labels.

Stage	Teacher action
35 to 45 minutes	Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

▼ Extension teaching cycles

Connected teaching cycle: Capstone models and baselines

Target: Fit and compare a **baseline** and a simple candidate

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: Compute the worked candidate accuracy, precision and recall.

Teacher check: Teacher explanation: Accuracy 16/20=80 percent; precision 4/6 and recall 4/6, both about 66.7 percent; FP 2 FN 2.

Individual check: For Capstone models and **baseline**s, use its worked case to evaluate this proposed explanation: “A final answer is enough”. Explain the deciding step.

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: Always predict clear from the 30/10 **training** counts; validation accuracy 14/20=70 percent.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Run the complete supplied investigation

> Keep predictions, observed or supplied outcomes and unmeasured cases separate.

Supplied inputs and checked guidance

Actual kit baseline and roles as supplied; final blanks are not negative labels. Keep derivedsourcegroups, features, versions and teacher review status. Independent bothmembersofeachpairmuststayone role, IDsnotfeatures; 24/8/8 pairswouldgive48/16/16 rows if justified, with class/coverage checks. No independent class counts or score supplied.

Illustrative baseline/candidate counts and original W4 copies remain separate practice.

Predict, investigate and record

1. 1. Write the hypothesis and expected **evidence** before running.
2. 2. Capstone Gate 1: peer-audit dataset card, leakage, split, **baseline** calculations, and **acceptance test**s; repair before build.
3. 3. Run one normal, one boundary, one missing, and one shifted case.
4. 4. Calculate the metric from raw counts or row-level errors.
5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Keep the experiment ledger

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome

Record	What to capture
Next test	One justified change and a new prediction

Diagnose the current misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record.

For this week: Compare a simple classifier with a **training**-majority **baseline** on the fictional parcel evaluation; no real handling or student decisions are supported.

A stronger claim makes a stronger project: Label the work as development or a demonstration, report the limitation and avoid an untouched-test claim; do not fabricate test data or results.

The new case can be copied from the example: The independent task changes the **evidence**.

Require a fresh explanation: Keep both members of each pair in one split, exclude pair ID from predictive **input**s, check group/class coverage and record split membership before **training**. Reserve final-test labels.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Extension misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: Always predict clear from the 30/10 **training** counts; validation accuracy 14/20=70 percent.

A stronger claim makes a stronger project: Model and pre**process**ing version, settings, feature order, split IDs, **prediction**/scoring rules, validation comparison, metric and **decision record**.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: **Baseline** 9/12=75 percent; candidate 10/12 about 83.3 percent; precision 2/3 and recall 2/3.

Candidate FP 1 FN 1 versus **baseline** FP 0 FN 3.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve the selected fresh80-row40-related-pair split repair for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

Diagnose and reteach

MISCONCEPTION TO REPAIR Capstone work should begin by **training** the most advanced available model.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Check these misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected experiment transfer

> Keep this key until after a first attempt; record any help and prior exposure.

Selected independent task and checked key

Selected case: A proposed 80-row dataset has 40 related pairs, and the team splits individual rows randomly then uses pair ID as a feature. Revise its plan.

Teacher explanation: Keep both members of each pair in one split, exclude pair ID from predictive **input**s, check group/class coverage and record split membership before **training**. Reserve final-test labels.

Judge trace, scope and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 State a question

Write a bounded question and a non-use statement for the supplied task.

Teacher explanation: Compare a simple classifier with a **training**-majority **baseline** on the fictional parcel evaluation; no real handling or student decisions are supported.

W2 Define fields

Explain why record ID, group ID and measured width have different roles.

Teacher explanation: Record ID tracks rows, group ID keeps related records together, and width is a potential measured feature available before **prediction**; IDs are not automatic **input**s.

W3 Set criteria

Name three acceptance criteria that do not require the candidate to win.

Teacher explanation: Examples: no group overlap, all planned **prediction**s accounted for, matched **baseline** comparison and documented limitations with denominators.

W5 Prepare a fallback

What should the team do if it cannot obtain a valid untouched final test?

Teacher explanation: Label the work as development or a demonstration, report the limitation and avoid an untouched-test claim; do not fabricate test data or results.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Assess and explain the answers

Worked case answer

Keep all views together in train/validation/test partitions. Define labels/use, fit a majority **baseline** on **training** labels and evaluate declared metrics.

Additional transfer practice question

Your capstone dataset includes duplicate photographs in both **training** and validation. What should be repaired before measuring the **baseline**?

Reasoned practice answer

Group duplicates or related examples so they do not cross the split; document the split and evaluate a **baseline** under the repaired protocol.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
</details>

Criterion	0 points	1 point	2 points
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Duplicate images in validation provide independent **evidence** because their filenames differ.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Keep paired views together: train=48 images, validation=16, test=16. Fit the majority label from **training** labels; choose metrics and model settings with **training**/validation.

Consult final labels only after selecting the system for its final evaluation.

Explanations for the additional workbook examples

W1 State a question

W2 Define fields

W3 Set criteria

W4 Apply independently

A proposed 80-row dataset has 40 related pairs, and the team splits individual rows randomly then uses pair ID as a feature. Revise its plan.

Teacher explanation: Keep both members of each pair in one split, exclude pair ID from predictive **input**s, check group/class coverage and record split membership before **training**. Reserve final-test labels.

W5 Prepare a fallback

Assess independent transfer

Selected case answer

Feedback that the learner uses

▼ **Extension questions and checked keys**

Capstone models and baselines — W1 Build the baseline

State the worked **training**-majority rule and validation accuracy.

Teacher explanation: Always predict clear from the 30/10 **training** counts; validation accuracy 14/20=70 percent.

Capstone models and baselines — W2 Evaluate the candidate

Compute the worked candidate accuracy, precision and recall.

Teacher explanation: Accuracy 16/20=80 percent; precision 4/6 and recall 4/6, both about 66.7 percent; FP 2 FN 2.

Capstone models and baselines — W3 Explain the tradeoff

Compare errors rather than only headline accuracy.

Teacher explanation: **Baseline** FP 0 FN 6; candidate FP 2 FN 2. Candidate finds four more blocked cases while falsely flagging two clear cases.

Capstone models and baselines — W4 Apply independently

Training 18 clear/6 blocked; validation **baseline**[[9, 0],[3, 0]], candidate[[8, 1],[1, 2]]. Calculate accuracy for each and candidate precision/recall.

Teacher explanation: **Baseline** 9/12=75 percent; candidate 10/12 about 83.3 percent; precision 2/3 and recall 2/3. Candidate FP 1 FN 1 versus **baseline** FP 0 FN 3.

Capstone models and baselines — W5 Freeze evidence

List what must be saved before the teacher releases final-test labels.

Teacher explanation: Model and pre**process**ing version, settings, feature order, split IDs, **prediction**/scoring rules, validation comparison, metric and **decision record**.

MAIN LESSON · SECTION 6

Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

Project evidence

Publish the frozen Plant Classifier design-and-data bundle with version IDs and approval record.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support and selected reassessment

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a **baseline**, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **requirement** and **acceptance criterion** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: Keep both members of each pair in one split, exclude pair ID from predictive **input**s, check group/class coverage and record split membership before **training**. Reserve final-test labels.

Fresh retry: three images of the same fictional parcel must stay in one split. Ask the learner to identify the group and record a split-level check before modelling.

Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Delayed return

Suggested return points in this level: ai-4/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 35: Capstone Train, Analyze, Improve. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week35:Capstone Train,Analyze,Improve. Carry the exact student kit or documented project data,**baseline** and source split into one controlled validation comparison. Freeze the chosen system before teacher-gated final reference release;Week36reports and defends the **evidence**.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

▼ Extension support and reassessment

Readiness and reassessment for Capstone models and baselines

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **baseline** and **prediction ledger** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: **Baseline** 9/12=75 percent; candidate 10/12 about 83.3 percent; precision 2/3 and recall 2/3. Candidate FP 1 FN 1 versus **baseline** FP 0 FN 3.

Fresh retry **baseline**[[8, 0],[4, 0]], candidate[[7, 1],[2, 2]]: accuracy 8/12 versus 9/12, candidate precision 2/3 and recall 2/4. Require the tradeoff and frozen settings.

Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

Planning and assessment reference

Teaching target: Define a feasible capstone question and acceptance criteria

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Explain why record ID, group ID and measured width have different roles.

Reasoned practice answer: Teacher explanation: Record ID tracks rows, group ID keeps related records together, and width is a potential measured feature available before **prediction**; IDs are not automatic **input**s.

Check every learner before discussion: Use the worked case for Capstone data and **requirement**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Ask for the relevant counts, units, procedure or saved record.

For this week: Compare a simple classifier with a **training**-majority **baseline** on the fictional parcel evaluation; no real handling or student decisions are supported.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: A derived crop of plantG is added to **training**. Require a **prediction** and a reason before comparing. Expected result: The final group separation is violated.

Reason: The crop retains the same source plant identity even with a new file name.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Earlier wording, alternatives and picture transcripts**

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

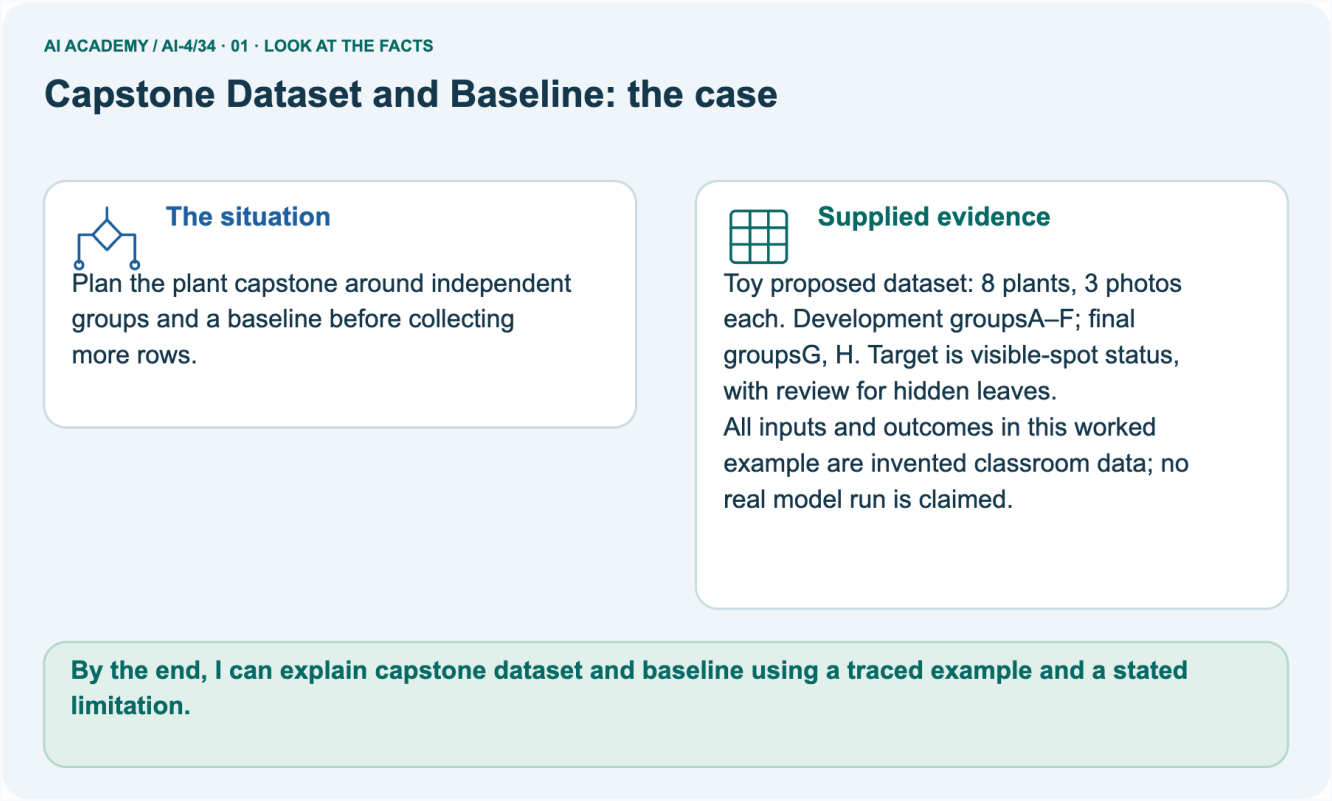
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

01. Capstone Dataset and Baseline: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

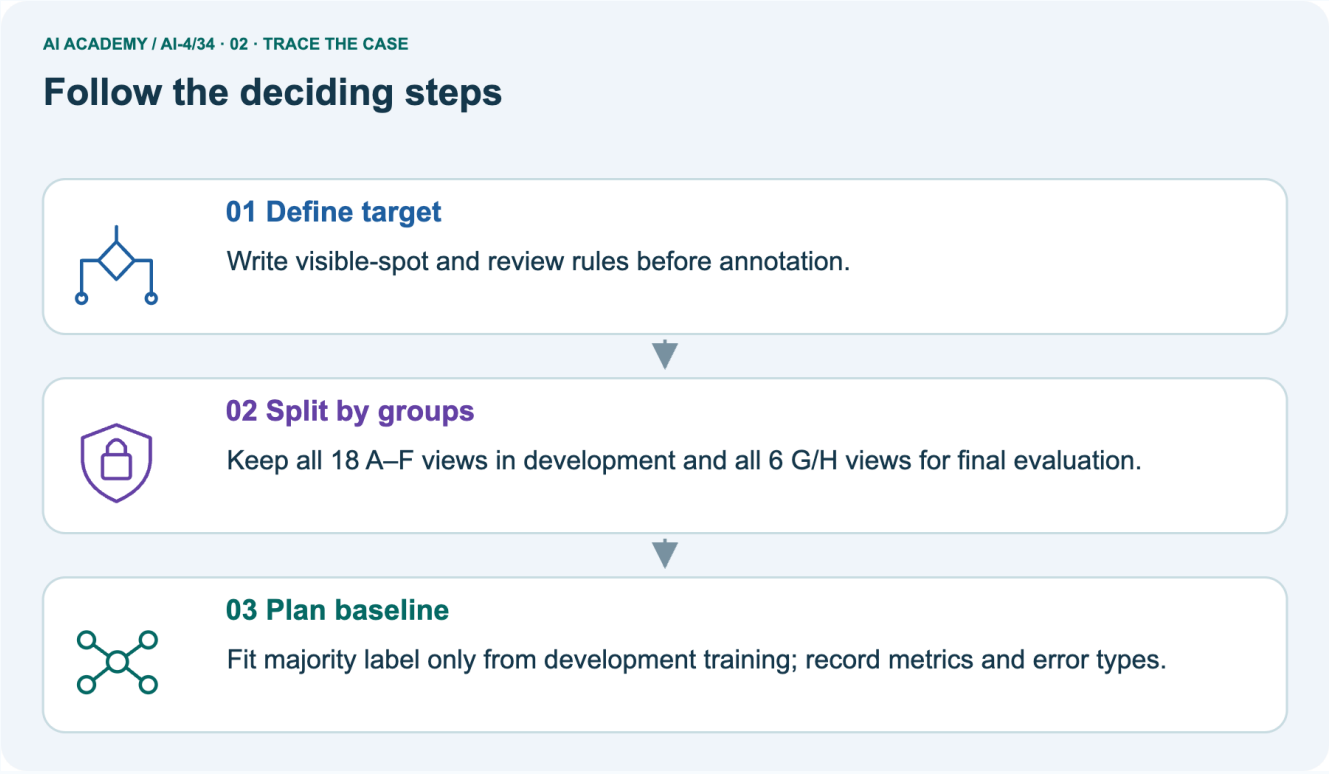
Plan the plant capstone around independent groups and a **baseline** before collecting more rows. Supplied facts: Toy proposed dataset: 8 plants, 3 photos each. Development groupsA–F; final groupsG, H.

Target is visible-spot status, with review for hidden leaves. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Define target: Write visible-spot and review rules before annotation. → Split by groups: Keep all 18 A–F views in development and all 6 G/H views for final evaluation. → Plan **baseline**: Fit majority label only from development **training**; record metrics and error types.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Four-step worked evidence for Week 34: Capstone Dataset and

Baseline

AI ACADEMY / AI-4/34 · 03 · INSPECT THE LESSON MODEL

Four-step worked evidence for Week 34: Capstone Dataset and Baseline

Step	Evidence and reasoning
1	Freeze dataset card v3 and grouped split s2 with no plant identity crossing sets.
2	Test set remains sealed; validation supports model choices.
3	Validation has 12 healthy and 8 spotted leaves. The always-healthy baseline scores 12/20=60% accuracy and 0/8=0% spotted recall. The rule has TP=5, FN=3, TN=10, FP=2: accuracy 75% and spotted recall 62.5%.
4	Acceptance requires valid citations, metric calculations, subgroup review, and safe escalation.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

Follow the labelled picture with the matching explanation. Open the vector diagram

Four-step worked **evidence** for Week 34: Capstone Dataset and **Baseline**. 1 | **Freeze** dataset card v3 and grouped split s2 with no plant identity crossing sets.. 2 | Test set remains sealed; validation supports model choices.. 3 | Validation has 12 healthy and 8 spotted leaves.

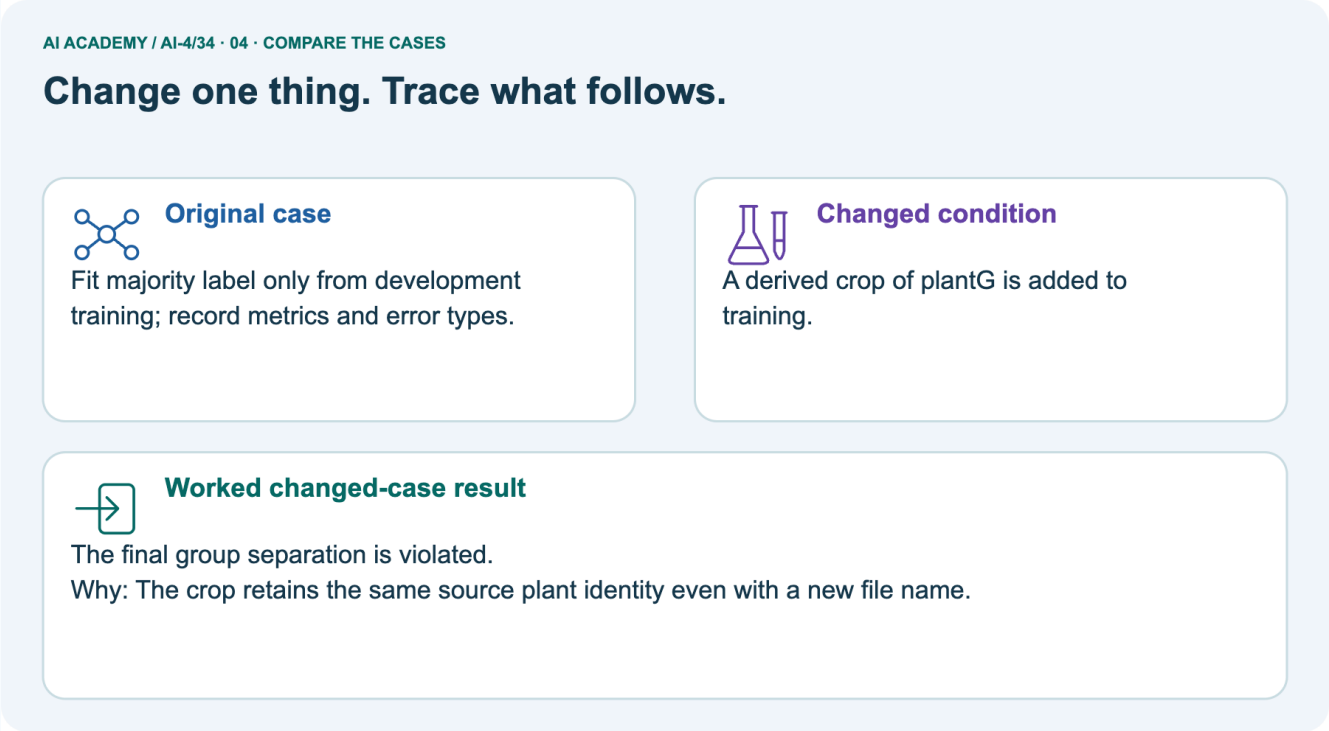
The always-healthy **baseline** scores 12/20=**60%** accuracy and 0/8=0% spotted recall.

The rule has TP=5, FN=3, TN=10, FP=2: accuracy 75% and spotted recall 62.5%.. 4 | Acceptance requires valid citations, metric calculations, subgroup review, and safe escalation.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Fit majority label only from development **training**; record metrics and error types. Changed condition: A derived crop of plantG is added to **training**. Expected result: The final group separation is violated.

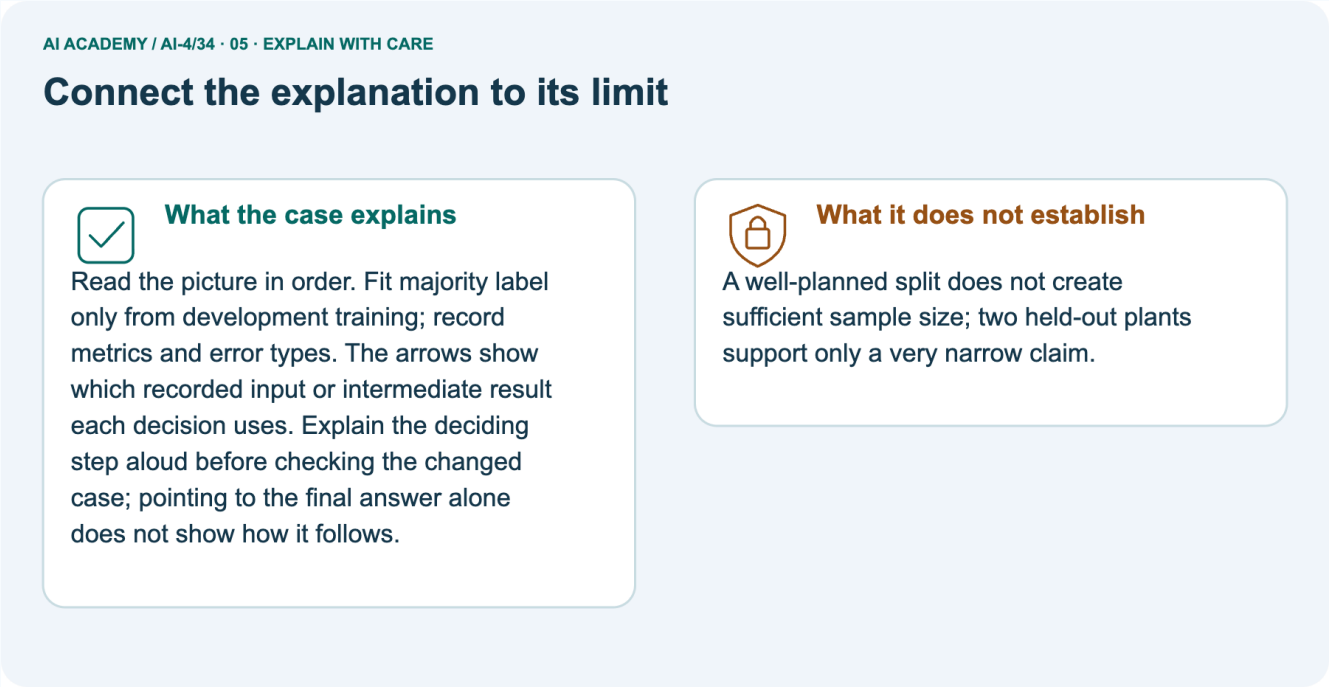
Reason: The crop retains the same source plant identity even with a new file name.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The crop retains the same source plant identity even with a new file name.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Fit majority label only from development **training**; record metrics and error types. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

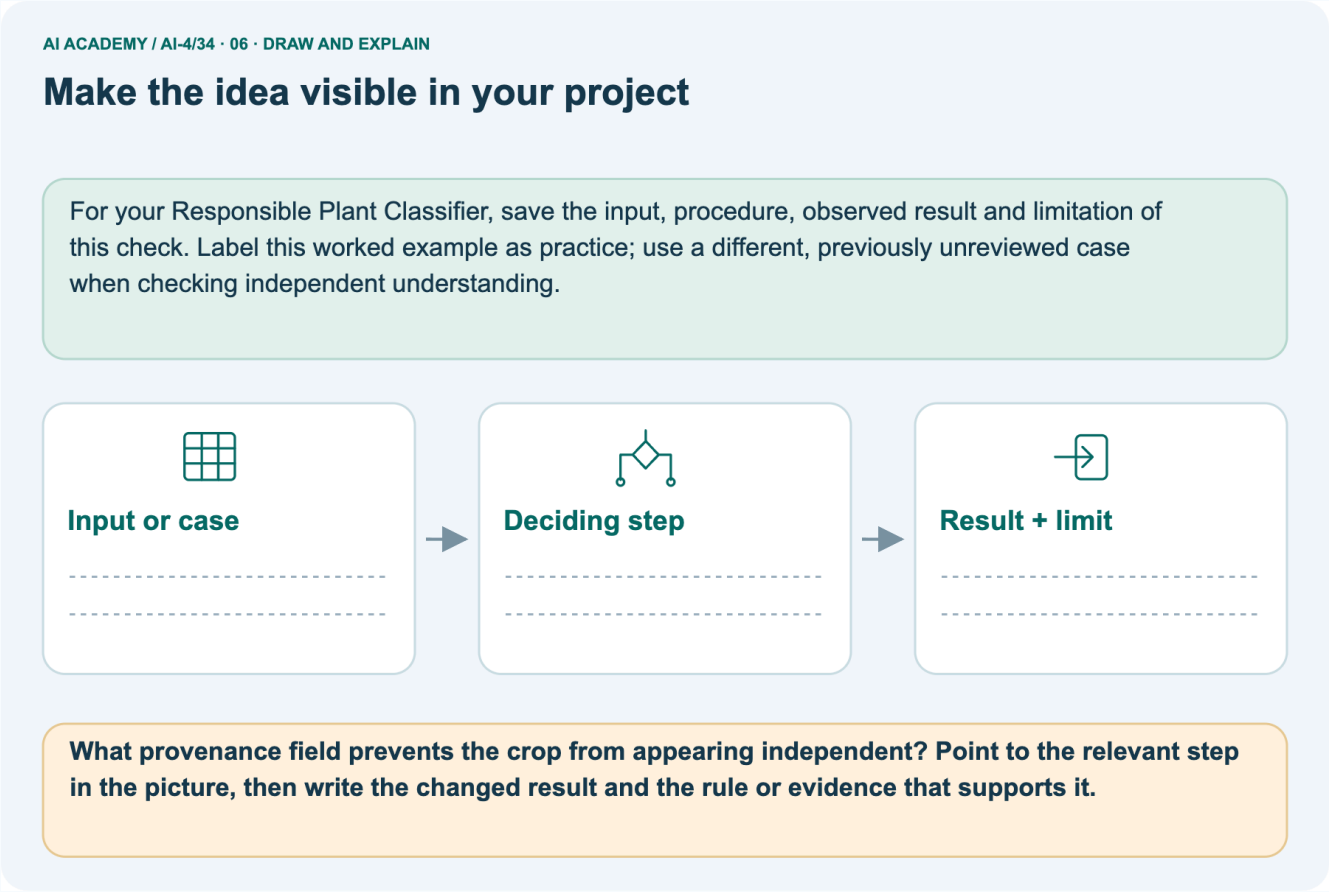
Limit: A well-planned split does not create sufficient sample size; two held-out plants support only a very narrow claim.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A well-planned split does not create sufficient sample size; two held-out plants support only a very narrow claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What provenance field prevents the crop from appearing independent? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Store original_plant_id=G and the parent image ID; all derived G records must remain with their assigned final group. Changed-case check: The final group separation is violated.

The crop retains the same source plant identity even with a new file name.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Capstone Dataset and Baseline: the case

02. Follow the deciding steps

03. Four-step worked evidence for Week 34: Capstone Dataset and Baseline

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.